

Dear Lightbridge Hiring Team,

I am writing to apply for the Senior Mechanical Engineer (Fuel Assembly Design) position. With a Ph.D. in Mechanical Engineering and over five years of R&D experience at Los Alamos National Laboratory developing multi-scale finite element models and surrogate simulations for nuclear fuel-cladding systems under extreme conditions, I am excited to contribute to the design, analysis, and licensing of Lightbridge Fuel assemblies.

At LANL I have built and validated advanced FEA workflows in MOOSE/BISON to evaluate the thermal-mechanical performance, irradiation creep, and thermal creep behavior of fuel-cladding systems (including zirconium-based alloys and advanced steels). This work involved implicit and explicit analyses in multi-scale modeling approach, polycrystal modeling, verification & validation against experimental results, uncertainty quantification, and the development of physics-based surrogates that bridge microscale insights to component-level predictions.

I am particularly drawn to Lightbridge's innovative metallic fuel technology and its potential to improve safety, performance, and economics in existing reactors and future SMRs. I believe my experience in nuclear fuel-cladding modeling, multi-physics FEA, irradiation/thermal creep, and physics-based surrogate modeling would allow me to contribute effectively to structural analysis of fuel assemblies as well as fabrication process simulations and licensing activities.

I would welcome the opportunity to discuss how my background aligns with Lightbridge's goals. Thank you for your consideration.

Warm regards,

Andre Ruybalid